SAI SHASHIDHAR POTANA

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years of experience building backend services and full-stack products. Since 2019, the work has been mostly in Python with Django, FastAPI, PostgreSQL, Redis, and AWS, at insurance startups where a small team owns a feature from the first product conversation to fixing it in production.

Backend work has centered on APIs, third-party data integrations, PDF document workflows, and background jobs for customer, agent, and internal products. Tests ship with the feature, code goes through review, and slow queries and endpoints get looked at after launch, not left for later. Comfortable building the React and TypeScript side when a feature needs it.

More recently, built production LLM features on AWS Bedrock, Google Vertex AI, OpenAI, and Anthropic, along with the evaluation pipelines and review tools used to test and improve them.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, HTML5, CSS3

Backend: Python, Django, FastAPI, REST APIs, GraphQL, WebSockets, AWS Lambda, API Gateway, Node.js

Data & Messaging: PostgreSQL, Redis, DynamoDB, SQS, EventBridge, Step Functions, query tuning, indexing, caching, background jobs

Cloud & Infrastructure: AWS, Terraform, AWS CDK, EC2, S3, EKS, Kubernetes, Docker, Cognito, CloudWatch

Testing & Observability: pytest, Jest, React Testing Library, Playwright, Cypress, Jasmine, Karma, CloudWatch logs, metrics and alarms, Bugsnag

AI / LLM: AWS Bedrock, Google Vertex AI, AutoML, OpenAI, Anthropic, Gemini, RAG, GraphRAG, function calling, structured prompting, agentic workflows, LLM evaluation, benchmark datasets, grading rubrics, regression evaluation

Frontend: React.js, Next.js 16, React Server Components, Redux Toolkit, Redux Saga, Zustand, TanStack Query, Material UI, Tailwind CSS, Storybook, design systems, accessibility

Product & Analytics: GTM, Hightouch, Amplitude, OpenReplay, A/B testing, event tracking

WORK EXPERIENCE

Telenav (Novo Insurance) | UT **March 2022 - Present**

Senior Software Engineer

- Built and shipped several new consumer, agent, AI, and internal products on a small team, taking each from rough requirements and technical design through launch and the fixes that followed. Backend services were in Python (Django and FastAPI) with PostgreSQL, Redis, and DynamoDB on AWS, and features were planned and tested alongside product managers, designers, QA, and frontend engineers.
- Integrated third-party data providers into our risk workflows, including Verisk, MVR (motor vehicle record) reports, and First Connect. Wrote the Python clients, response validation, retries, and fallback rules so a slow or failed vendor call did not break the customer's flow.
- Found and fixed slow API endpoints by rewriting inefficient PostgreSQL queries, adding indexes, removing repeated database calls, and caching frequently read data in Redis with expiry rules that matched how often it changed.
- Built backend document workflows that generate and export PDFs from application data, including documents drafted with LLM help and checked against business rules before they went out.
- Moved heavy and long-running work off the request path into background jobs on SQS, EventBridge, Step Functions, and Lambda, with infrastructure defined in Terraform and AWS CDK.
- Wrote pytest unit and integration tests for APIs and business rules, reviewed teammates' pull requests, and set up CloudWatch logs, metrics, and alarms so failed jobs and slow endpoints were visible early.

- Built LLM features with AWS Bedrock, Google Vertex AI, OpenAI, Anthropic, and Gemini for customer-service analysis, requirements generation, and internal tools, using RAG, GraphRAG, function calling, and backend tool calls.
- Set up the evaluation process for those features: curated datasets of real business scenarios and edge cases, wrote grading rubrics, and ran repeated evaluations across prompts, models, and retrieval strategies as a batch pipeline. When results got worse, traced the failure through the prompt, retrieved context, tool calls, and business rules to find the real cause.
- Built internal apps that customer-service reps, agents, and engineers use to manage operational work, inspect application data, review AI output, and troubleshoot production issues.
- Maintained a shared React/TypeScript component library published to npm and documented in Storybook, added analytics and A/B testing with GTM, Hightouch, Amplitude, and OpenReplay, and raised Lighthouse performance from about 40 to 90.

Motion Auto | UT **June 2019 - March 2022**

Software Engineer

- Built backend APIs and services in Python on AWS Lambda, API Gateway, Step Functions, and PostgreSQL for insurance, claims, roadside assistance, customer accounts, and authentication.
- Connected customer and agent applications to internal platforms and third-party systems through REST and GraphQL APIs.
- Built responsive, data-heavy React front ends with Redux, Hooks, Context, and Material UI, and moved legacy class-based components to functional components and Hooks to make them easier to test and reuse.
- Implemented login and access control with AWS Cognito, SSO, protected routes, and authenticated API calls.
- Worked with product, design, and QA to turn requirements into tested features and to track down customer-facing issues.

Intellitide | TX **October 2018 - June 2019**

Software Engineer

- Developed a warehouse-management application using Angular 7, TypeScript, RxJS, Node.js, REST APIs, AWS API Gateway, and Cognito, covering inventory management, order tracking, authentication, and internal operations.
- Built reusable Angular components, services, and Node.js backend APIs, and improved load times with lazy loading, Ahead-of-Time compilation, and tree shaking.
- Wrote unit tests with Jasmine and Karma and worked with engineering and product to ship and maintain production features.

EDUCATION

Master of Science, Computer Engineering | University of Houston-Clear Lake, Houston, TX